EE2-21

SOLUTIONS: Feedback Systems

1. a) i) The transfer function for the circuit in Figure 1.2 is given by

$$\frac{Z_f(s)}{Z_i(s)} = -\frac{C_i(s+1/R_iC_i)}{C_f(s+1/R_fC_f)} = -\frac{C_is+1/R_i}{C_fs+1/R_f}$$

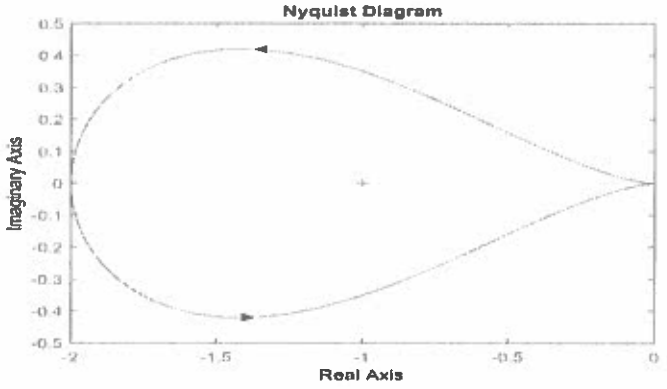
Putting in the values: $\boxed{G(s) = G_1(s)G_2(s) = \frac{1}{(s+1)(s+2)}}$.

ii) The DC gain of $G(s)$ is therefore $\boxed{G(0) = 0.5.}$

iii) $\boxed{y_{ss} := \lim_{t\to\infty} y(t) = \lim_{s\to 0} sy(s) = \lim_{s\to 0} sG(s)u(s) = G(0) = 0.5.}$

iv) $\lim_{t\to\infty} e(t) = \lim_{s\to 0} se(s) = \lim_{s\to 0} \frac{sr(s)}{1+K_pG(s)} = \frac{1}{1+K_pG(0)} \leq .01 \Rightarrow \boxed{K_p \geq 198.}$

b) i) The Nyquist diagram is shown below.



ii) The Nyquist criterion: $N = Z - P$ where N is the number of clockwise encirclements of $-1/K_p$, Z is the number of unstable closed loop poles and P is the number of unstable open loop poles ($= 1$). So for:

- $0 < K_p < 0.5$, $N = 0$ and so $Z = 1$,
- $0.5 < K_p < \infty$, $N = -1$ and so $Z = 0$
- $-\infty < K_p < 0$, $N = 0$ and so $Z = 1$.
- when $K_p = 0.5$ the closed loop is marginally stable.

iii) For $K_p = 1$, the closed loop is stable. The loop gain can be increased without bound and reduced by up to 50% without losing stability.

iv) The closed loop transfer function, DC gain and damping ratio are

$$\boxed{H(s) = \frac{4K_p}{s^2+s+2(2K_p-1)}}, \quad \boxed{H(0) = \frac{2K_p}{2K_p-1}}, \quad \boxed{\zeta = \frac{1}{2\sqrt{2(2K_p-1)}}}.$$

When $r(t)$ is a unit step, good steady–state response requires $H(0) \sim 1$, or equivalently, large K_p. However, large K_p will result in small ζ, which results in large overshoot and an oscillatory response.

2. a) The Nyquist diagram is shown in the next page. The real–axis intercepts can be found from the Routh arrray. The characteristic equation is

$$1+K_pG(s)=0 \Rightarrow s^2+(3-K_p)s+2+3K_p=0$$

The Routh array

$$\begin{array}{c|cc} s^2 & 1 & 2+3K_p \\ s^1 & 3-K_p & \\ 1 & 2+3K_p & \end{array}$$

The values $K_p=3$ and $K_p=-2/3$ result in a zero row, and so the real–axis intercepts are obtained as $-1/K_p$ and the corresponding frequencies are obtained from the auxiliary polynomials and are $\boxed{-1/3,\ \omega=\sqrt{11}}$ and $\boxed{3/2,\ \omega=0.}$

b) The $\boxed{\text{gain margin}=3}$ from the Routh array. For the phase margin, we first find the real frequency ω such that $|G(j\omega)|=1$, or

$$9+\omega^2=(1+\omega^2)(4+\omega^2) \Leftrightarrow \omega^4+4\omega^2-5=0 \Leftrightarrow (\omega^2-1)(\omega^2+5)=0 \Rightarrow \omega=\pm 1.$$

Therefore the cross–over frequency $\boxed{\omega_c=1}$ and the angle of $G(j\omega_c)=-j$ is $-90°$ and so the $\boxed{\text{phase margin}=90°.}$ The stability margins are adequate so $\boxed{\text{the design is robust against uncertainties in the gain and phase of } G(s).}$ Furthermore, $\boxed{\text{we expect the transient response to be acceptable}}$ since the Nyquist plot is not too close to the point -1.

c) When $K(s)=1$, the closed loop transfer function is

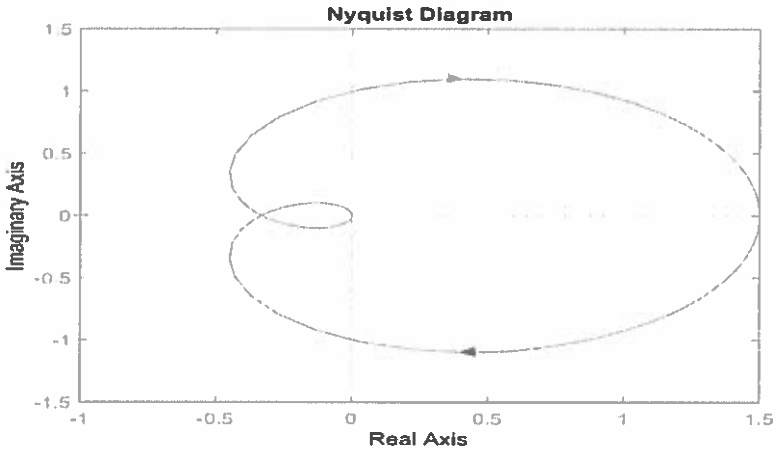
$$H(s)=\frac{G(s)}{1+G(s)}=\frac{3-s}{s^2+2s+5} \Rightarrow H(0)=0.6.$$

Therefore, $\boxed{\text{the steady–state output is 0.6.}}$ Since the reference signal is 1, $\boxed{\text{the steady–state response of the system is not adequate.}}$

d) The Nyquist criterion: $N=Z-P$ where N is the number of clockwise encirclements of $-1/K_p$, Z is the number of unstable closed loop poles and P is the number of open loop poles ($=0$). So for

- $0<K_p<3$, $N=0$ and so $Z=0$,
- $3<K_p<\infty$, $N=2$ and so $Z=2$,
- $-\infty<K_p<-2/3$, $N=1$ and so $Z=1$,
- $-2/3<K_p<0$, $N=0$ and so $Z=0$,
- when $K_p=3$ and $K_p=-2/3$ the closed loop is marginally stable.

e) Note from Parts b and c that the transient response is adequate while the steady–state response is not. Hence $\boxed{\text{we need to improve the steady–state performance.}}$ Since phase–lag compensation increases low frequency gain, and so improves steady–state tracking it follows that $\boxed{\text{the system requires phase–lag compensation.}}$


Nyquist Diagram
Imaginary Axis
Real Axis
1.5
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1
1.5

3. a) For a critically damped response with a settling time of approximately 1 s, the poles must be repeated and located at $\boxed{p_1 = p_2 = -4.}$

b) The closed–loop characteristic equation is $1 + K_p G(s) = 0$ or

$$\boxed{s^2 + 2s + K_p(s + 8/3) = s^2 + (2 + K_p)s + 8K_p/3 = 0.}$$

c) The Routh array is given by

$$\begin{array}{c|ll} s^2 & 1 & 8K_p/3 \\ s & 2+K_p & \\ 1 & 8K_p/3 & \end{array}$$

Thus for closed–loop stability, $K_p > 0$ and $K_p > -2$, or $\boxed{K_p > 0.}$

d) To achieve the design specifications, $s_1 = s_2 = -4$ must be roots of the characteristic equation. It follows that we must have

$$s^2 + (2 + K_p)s + 8K_p/3 = (s+4)^2 = s^2 + 8s + 16.$$

Therefore $\boxed{K_p = 6.}$

e) Here $r(s) = 1/s^2$ and $e(s) = r(s)/(1 + 6G(s))$. Using the final value theorem of the Laplace transform:

$$\boxed{\lim_{t\to\infty} e(t) = \lim_{s\to 0} se(s) = 1/8.}$$